



AeroBlade User Guide

Steady HAWT rotor aerodynamics, design, optimization and CAD export

Software	AeroBlade 2.1.3
Document	AB-UG-2.1.3
Status	Versioned software documentation
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License	Proprietary academic/research license; commercial use requires written authorization

AeroBlade 2.1.3 is distributed as a Windows portable application. Verify the SHA-256 checksum before scientific use.

Document control

2.1.3	2 August 2026	Portable installation, scientific workflow, licensing and support	Version 2.1.3

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1. Purpose and declared scope

AeroBlade is a desktop engineering application for preliminary steady, axial-flow horizontal-axis wind-turbine (HAWT) rotor analysis. It combines airfoil geometry and polar handling, blade-element/momentum (BEM) calculations, regulated power and annual-energy estimates, inverse design, numerical optimization and aerodynamic blade-surface export.

Engineering boundary. AeroBlade is research software. It is not a certification, structural, fatigue, aeroelastic, controls, noise, turbulent-wake or safety-load tool. Results require mesh/iteration sensitivity checks and comparison with applicable experiments or higher-fidelity methods.

1.1 Intended users

- Researchers comparing steady BEM formulations and empirical corrections.
- Teachers and students learning rotor aerodynamics and airfoil-polar workflows.
- Early-stage designers screening rotor dimensions, blade distributions and energy sensitivity.
- Users preparing aerodynamic surface geometry for downstream CAD/CFD work.

1.2 Model boundary

Included	Not represented
Steady axial BEM; tip/root losses; high-induction correction	Yawed flow, dynamic inflow, dynamic stall
Single or Reynolds-family 2-D polars; Viterna extrapolation	3-D CFD, free-vortex wake, tower interference
Rotational-augmentation sensitivity models	Aeroelastic deformation and structural loads
Power curve, Weibull AEP and simple losses	Controller dynamics, grid model and financial assessment
Closed aerodynamic STL surface	Manufacturing-ready solid model, laminates, root inserts

2. Portable installation and integrity check

1. Download the ZIP and its adjacent .sha256 file from the official GitHub Release or DOI record.
2. Keep the ZIP intact until its SHA-256 value has been checked.
3. Extract the complete AeroBlade folder to a writable local path. Do not run the EXE from inside the ZIP.
4. Start AeroBlade.exe. No separate Python installation is required.
5. Keep the _internal folder beside AeroBlade.exe; moving only the EXE breaks the application.

2.1 SHA-256 check in Windows PowerShell

```
Get-FileHash ".\AeroBlade-2.1.3-Windows-x64-Portable.zip" -Algorithm SHA256
```

Compare the displayed hexadecimal value with the value in the .sha256 file. A mismatch means the archive must not be used. RELEASE-MANIFEST.json provides component hashes after extraction.

2.2 First start, data and uninstall

- Windows may display a reputation warning because the release is not yet code-signed. Confirm that the checksum and download origin are correct before selecting Run anyway.
- %LOCALAPPDATA%\AeroBlade stores preferences and logs; projects remain wherever the user saves them.
- Help > Open log folder opens the diagnostic directory.
- Uninstall by deleting the extracted folder. Optionally remove %LOCALAPPDATA%\AeroBlade after preserving required logs.

3. Interface map

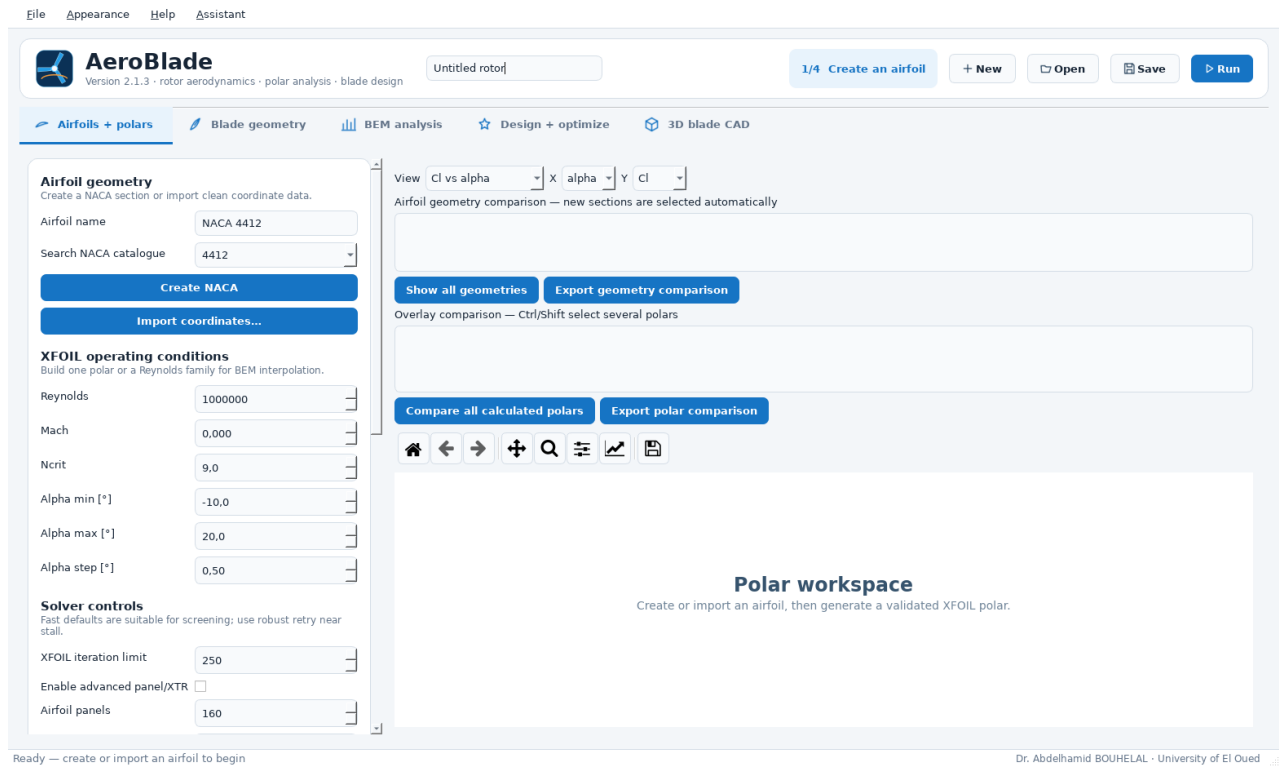


Figure 1. AeroBlade workspace and Airfoils + polars tab.

Area	Function
Header	Project name, workflow cue, New/Open/Save/Run shortcuts.
Five tabs	Airfoils + polars; Blade geometry; BEM analysis; Design + optimize; 3D blade CAD.
File	Project actions and export of the currently displayed variables.
Appearance	Theme and accent controls persisted between sessions.
Help	User guide, V&V report, license, notices, log folder and About dialog.
Plot toolbar	Home, navigation, pan, zoom, subplot settings and image export.

4. Recommended end-to-end workflow

1. Create or import each airfoil geometry and inspect orientation, closure and normalization.
2. Generate/import polars spanning the local Reynolds-number and angle-of-attack envelope; extrapolate to 360 degrees where stall or shutdown is possible.
3. Create/import blade stations and assign a polar key at every aerodynamic station.
4. Select operating conditions and a documented BEM model set; solve one operating point first.
5. Inspect convergence, induction factors, loads and Reynolds coverage before accepting integrated coefficients.
6. Run radial-station and convergence-parameter sensitivity studies.
7. Generate the regulated power curve and AEP only after operating-point checks pass.
8. Use inverse design/optimization for candidate geometry, then revalidate the whole operating envelope.
9. Export plots, CSV data, the project and optional STL together with assumptions and the software version.

Reproducibility rule. A scientific result should travel with the .absproj project, input polar sources, exported CSV data, AeroBlade version, model/correction choices and release checksum.

5. Airfoils and polars

5.1 Geometry creation and import

The catalogue creates standard NACA four- and five-digit airfoils. NACA six-series and proprietary/tabulated profiles must be imported from authoritative x/c , y/c coordinates. Imported coordinates may be whitespace- or comma-separated. Check that chord normalization, point order and trailing-edge treatment match the intended profile. Duplicate names receive a suffix instead of silently replacing data.

5.2 XFOIL polar generation

- Choose Reynolds number, Mach number, Ncrit, transition locations, angle interval, increment and maximum viscous iterations.
- Generate several Reynolds cases spanning the blade rather than relying on a single polar.
- Enable retry of unconverged angles when appropriate; missing points remain missing if XFOIL does not converge.
- Inspect requested versus returned alpha bounds and do not interpret an interrupted polar as complete.
- The bundled XFOIL 6.99 process is independent third-party software; its GPL and source archive are supplied under LICENSES.

5.3 Polar import, inspection and extrapolation

Imported polar tables require alpha, Cl and Cd; an optional fourth column is interpreted as Cm. Plot lift, drag, moment, efficiency or arbitrary X-Y combinations. Viterna-Corrigan extrapolation supplies a two-sided 360-degree continuation; it does not turn sparse or poor-quality measurements into validated post-stall data. Preserve the measured interval and record aspect-ratio/CDmax assumptions.

6. Blade geometry

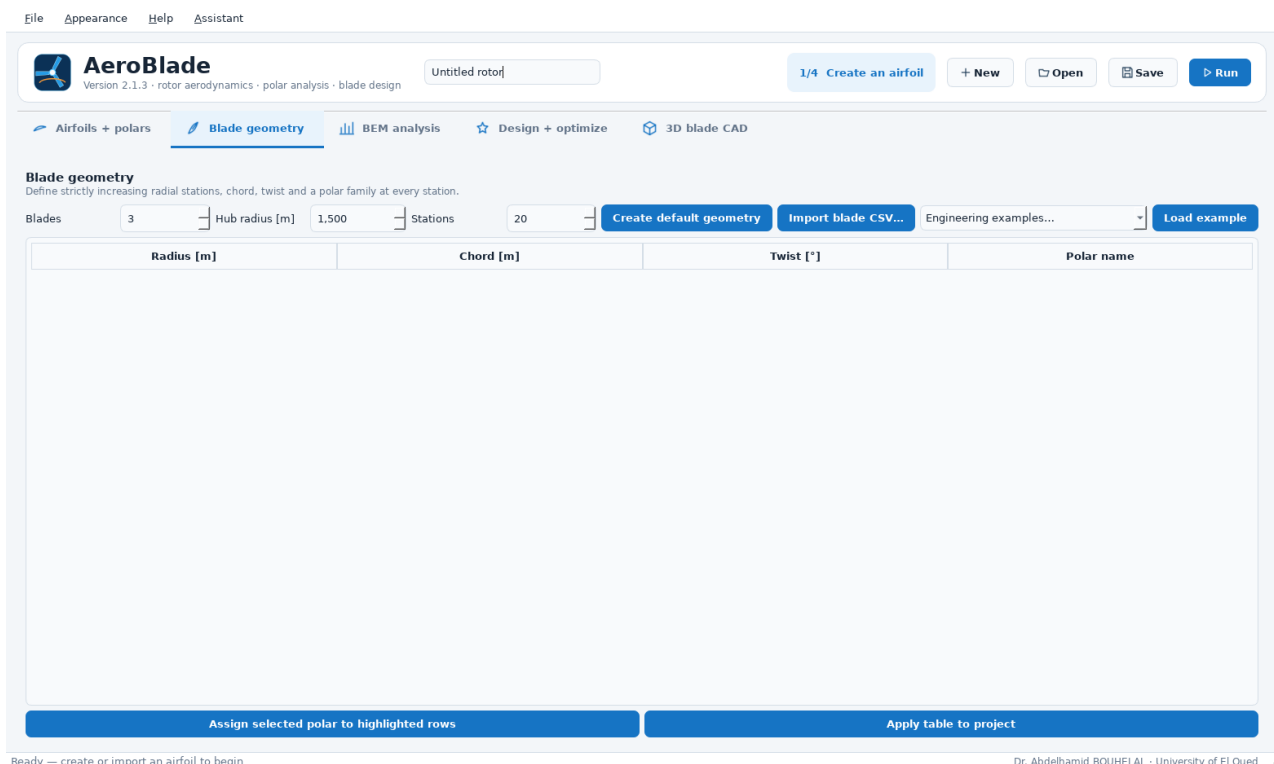


Figure 2. Blade station editor and geometry visualization.

Each row defines radius [m], chord [m], twist [deg] and a polar key. Radius must increase monotonically; chord must remain positive; and every active section must resolve to an available polar or Reynolds family. Interpolation occurs between stations during BEM discretization.

Check	Recommended evidence
Units and radius definition	Confirm dimensional metres and the same rotor radius used in performance normalization.
Hub/root region	Define the aerodynamic start radius; do not invent lifting data inside the structural root.

Check	Recommended evidence
Polar assignment	Document airfoil transition stations and Reynolds database coverage.
Station sensitivity	Repeat the calculation with at least one finer radial discretization.
Built-in benchmarks	Phase II/VI entries provide geometry only; supply authoritative S809 polars separately.

6.1 Blade CSV schema

radius_m, chord_m, twist_deg, polar

7. BEM operating-point analysis

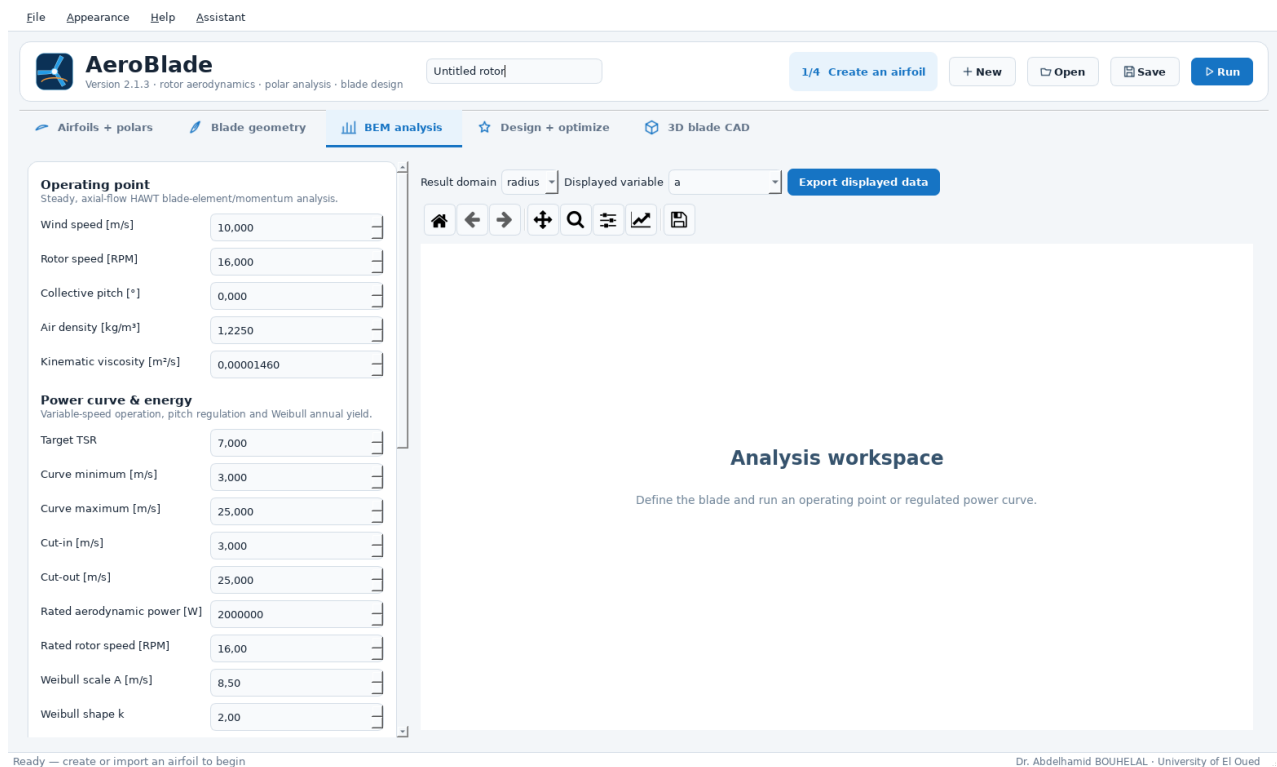


Figure 3. BEM operating conditions, corrections and result workspace.

Set wind speed, rotor speed, collective pitch, air density and kinematic viscosity. The solver resolves axial and tangential induction at each radial element, evaluates local Reynolds number and interpolates the assigned polar family. Integrated thrust, torque and power follow from section loads.

7.1 Model selections

Model group	Use and reporting requirement
Prandtl tip/root loss	Momentum-loss factor for finite blades; report whether tip and root are enabled.
High induction	Buhl or DTU/Madsen relation; include the selected branch in methods.
Rotational augmentation	None, Snel or Himmelskamp sensitivity; empirical and configuration-dependent.
QBlade-style local factor	Section-load correction reported separately; does not imply solver identity with QBlade.
Reynolds treatment	Family interpolation is preferred; optional drag scaling is a sensitivity approximation.

7.2 Acceptance checks for one operating point

- All active stations converged; investigate any clipped or non-converged station.
- Local Reynolds numbers lie within, or are defensibly close to, the polar-family envelope.
- Induction and loss factors vary physically along the span without isolated discontinuities.
- Power equals torque times angular speed within numerical precision.
- C_p remains below the Betz limit; an unusually high value normally signals input, sign or convergence errors.
- Integrated outputs remain acceptably stable under a finer radial grid and tighter tolerance.

8. Power curve and annual energy

Define target tip-speed ratio, curve limits, cut-in/cut-out, rated rotor speed, pitch regulation and rated aerodynamic power. The curve applies drivetrain efficiency, fixed electrical losses and availability for the reported energy estimate. Weibull scale A and shape k define the illustrative wind distribution.

Interpretation. The GUI AEP is only as representative as the entered site distribution and loss assumptions. It is not a bankable energy-yield assessment. Extend the wind-speed range when probability coverage is below 98%.

- Inspect power, torque, C_p , C_t , rotor speed, pitch and convergence across the full wind range.
- Check transitions at cut-in, rated operation and cut-out for physically defensible discontinuities.
- Report whether rated power refers to aerodynamic shaft power or post-loss electrical power.
- Export the displayed curve to CSV and retain Weibull/loss inputs with the project.

9. Design, optimization and CAD

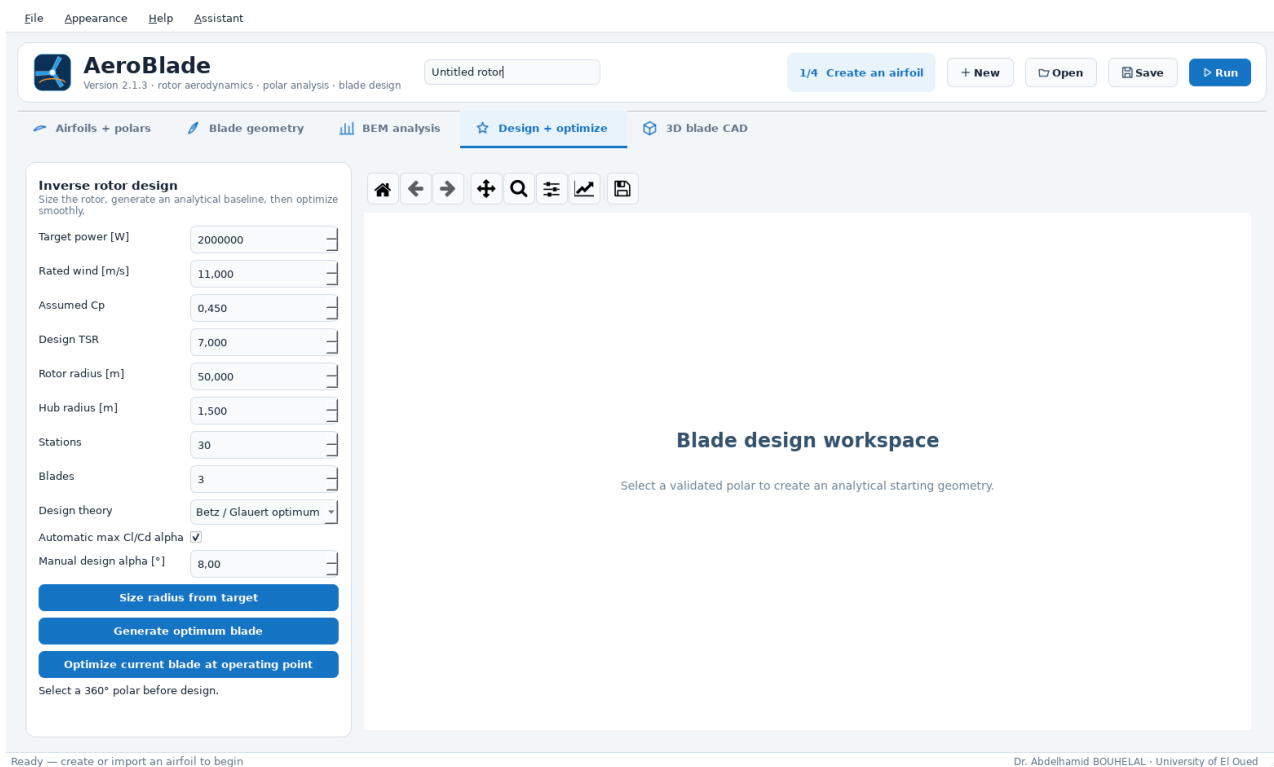


Figure 4. Rotor sizing, inverse design and numerical optimization controls.

Inverse-design tools create Glauert or Schmitz starting distributions from an idealized design point. Numerical optimization then modifies selected variables within user-defined bounds. A numerical optimum is not automatically a manufacturable, structurally safe or off-design-robust blade. Save the baseline and report bounds, objective, iterations and final constraint checks.

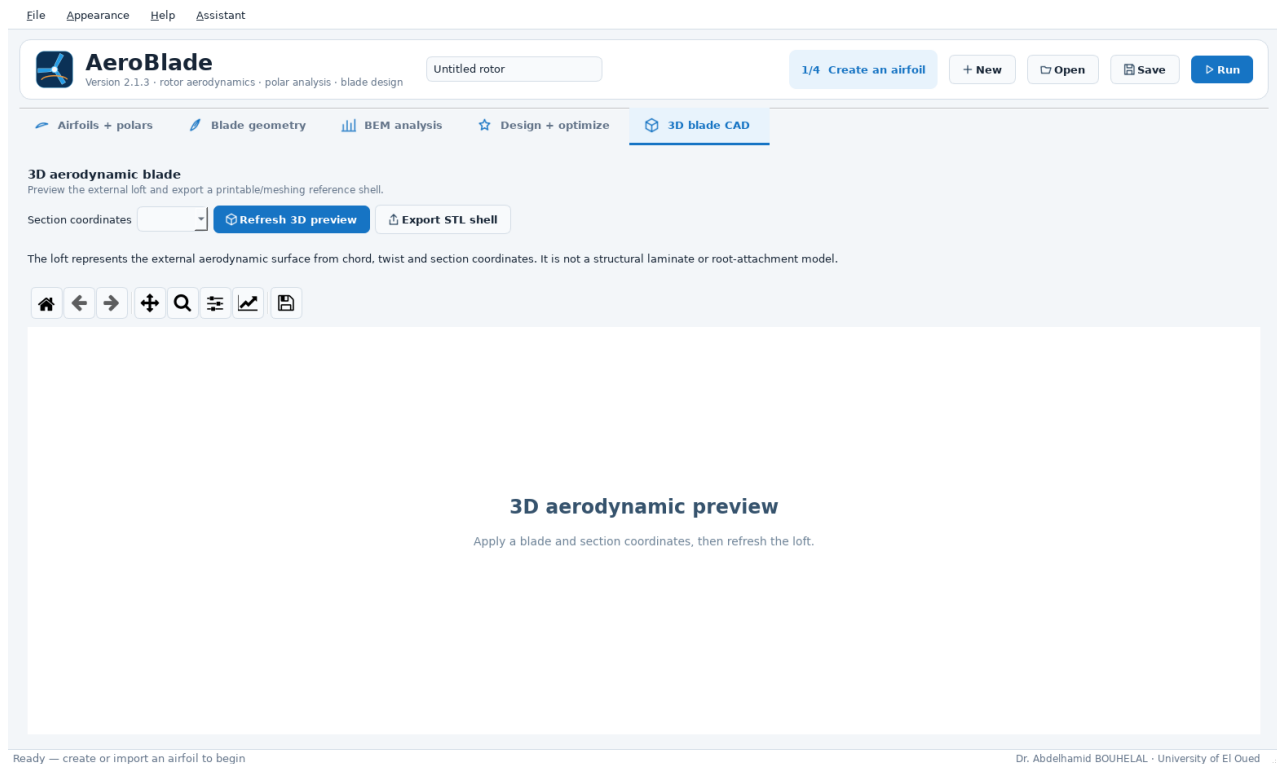


Figure 5. Aerodynamic blade-surface preview and STL export.

The 3D tab lofts assigned airfoil sections using chord and twist and closes root and tip for a watertight aerodynamic shell. Exported ASCII STL has no structural webs, spar caps, laminates, lightning protection, bolts or root inserts. Inspect topology and coordinate conventions before importing into CAD or CFD.

10. Projects, exports and scientific records

Artifact	Purpose
.absproj	JSON project containing coordinates, polars, blade geometry and settings.
CSV export	Exact variables currently displayed; suitable for independent plotting and analysis.
PNG/SVG/PDF plot	Presentation artifact; use CSV as the quantitative record.
STL	Aerodynamic surface exchange; not a structural model.
aeroblade.log	Diagnostic record for failures and support requests.
RELEASE-MANIFEST.json	Release version, component hashes, dependency versions and automated gate results.

10.1 Minimum methods statement

A publication should identify AeroBlade version and DOI, release checksum, rotor geometry source, airfoil/polar sources and Reynolds ranges, BEM loss/high-induction/rotational models, radial stations, convergence tolerance and relaxation, operating conditions, power-curve regulation and all energy-loss assumptions.

11. Troubleshooting

Symptom	Likely cause and action
Application does not start	Extract the full folder; retain _internal; inspect Help > Open log folder or %LOCALAPPDATA%\AeroBlade\logs.
XFOIL returns few points	Increase iterations, reduce alpha step/range, review Ncrit/transition/Mach, normalize geometry and use retry.
BEM station does not converge	Inspect polar coverage and discontinuities; adjust discretization, tolerance/relaxation and model selection.
Cp is implausible	Check units, radius/chord/twist signs, polar assignment, rotor speed, pitch and convergence.
AEP probability coverage is low	Extend the power-curve range and verify Weibull A/k and cut-out.
STL import looks twisted or scaled	Confirm metre units, coordinate system, twist sign and downstream import units.

12. Quick start in French / Démarrage rapide

1. Vérifier le SHA-256 du ZIP, puis extraire tout le dossier; ne pas déplacer AeroBlade.exe seul.
2. Créer ou importer les profils et générer plusieurs polaires couvrant les nombres de Reynolds de la pale.
3. Définir les stations rayon-corde-vrillage et affecter une polaire à chaque station.
4. Régler la vitesse du vent, la rotation, le pas, la densité et la viscosité, puis choisir les corrections BEM à documenter.
5. Calculer d'abord un point de fonctionnement; vérifier la convergence de toutes les stations, Cp, les facteurs d'induction et la couverture Reynolds.
6. Répéter avec un maillage radial plus fin et une tolérance plus stricte avant d'accepter les résultats.
7. Calculer ensuite la courbe de puissance/AEP et exporter le projet, les CSV, les figures et les hypothèses.
8. Le STL est une enveloppe aérodynamique; il ne constitue pas une conception structurelle certifiée.

13. License, citation and support

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